

A COMBINATORIAL INTERPRETATION OF DÉSARMÉNIEN'S FORMULA

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ABSTRACT. Using decorated permutations and Ji's sign-reversing involution, we give a combinatorial interpretation of Désarménien's formula for the distribution of the reduced major index over derangements. In particular, we show that the main families contribute uniformly to all residue classes modulo n , while a unique exceptional object produces the correction term.

1. INTRODUCTION

Let $\mathfrak{S}_n$ denote the symmetric group of permutations of the set $[n] = \{1, 2, \dots, n\}$. We call $i \in [n]$ a *fixed point* of $\sigma \in \mathfrak{S}_n$ if $\sigma_i = i$. A permutation $\sigma \in \mathfrak{S}_n$ with no fixed points is called a *derangement*. Denote by $\mathcal{D}_n$ the set of derangements in $\mathfrak{S}_n$. Let d_n be the cardinality of $\mathcal{D}_n$. By counting permutations according to their number of fixed points and using binomial inversion, one obtains

$$d_n = \sum_{k=0}^n (-1)^k \frac{n!}{k!}, \quad n \geq 0. \quad (1.1)$$

It follows that the derangement numbers d_n satisfy the recurrence

$$d_0 = 1 \text{ and } d_n = nd_{n-1} + (-1)^n, \quad n \geq 1. \quad (1.2)$$

Several combinatorial proofs of the recurrence (1.2) are known; see, for instance, the works of Désarménien [Dés84], Remmel [Rem83], Wilf [Wil84], Benjamin and Ornstein [BO17], and Elizalde [Eli21].

For a permutation $\sigma = \sigma_1\sigma_2 \cdots \sigma_n \in \mathfrak{S}_n$, an index $i \in [n-1]$ is called a *descent* of σ if $\sigma_i > \sigma_{i+1}$. The descent set and descent number of σ are defined by, respectively,

$$\text{Des}(\sigma) := \{i \in [n-1] : \sigma_i > \sigma_{i+1}\} \text{ and } \text{des}(\sigma) := |\text{Des}(\sigma)|.$$

The *major index* of σ is defined as the sum of its descents, i.e.,

$$\text{maj}(\sigma) := \sum_{i \in \text{Des}(\sigma)} i.$$

MacMahon [Mac60] proved the elegant identity

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{maj}(\sigma)} = [n]_q! = [1]_q [2]_q \cdots [n]_q. \quad (1.3)$$

Here $[0]_q! := 1$, and for $n \geq 1$,

$$[n]_q := \frac{1 - q^n}{1 - q} = 1 + q + \cdots + q^{n-1}.$$

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The *q-derangement numbers* are defined by $d_0(q) = 1$ and for $n \geq 1$,

$$d_n(q) := \sum_{\sigma \in \mathcal{D}_n} q^{\text{maj}(\sigma)}. \quad (1.4)$$

The following formula (1.5) was first obtained by Gessel in unpublished work on the joint distribution of descents, the major index, and cycle structure. These ideas were subsequently developed in the work of Gessel and Reutenauer [GR93] on permutations with prescribed cycle structure and descent set. Wachs [Wac89] gave a combinatorial derivation that closely parallels the classical inclusion–exclusion proof for derangements.

Theorem 1.1 (Gessel–Reutenauer–Wachs). *For $n \geq 1$, we have*

$$d_n(q) = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \frac{[n]_q!}{[k]_q!}. \quad (1.5)$$

Postnikov [Pos06] introduced the notion of decorated permutations, that is, permutations whose fixed points are assigned one of two signs. More precisely, each fixed point i may appear either as an ordinary fixed point i or as a signed fixed point $\bar{i}$. Denote by E_n the set of decorated permutations of $[n]$.

Recently, Ji [Ji26] used decorated permutations to give a direct combinatorial proof of (1.5), which answered a question raised by William Y.C. Chen. Her proof has two main ingredients. First, she establishes a major-index generating function for decorated permutations with a prescribed number of signed fixed points. Second, she constructs a sign-reversing and descent-set-preserving involution on decorated permutations having at least one fixed point. After cancelation, the surviving objects are precisely the ordinary derangements, which yield the Gessel–Reutenauer–Wachs formula (1.5) directly.

From the preceding Theorem 1.1, one can easily derive the following result.

Proposition 1.2. *The q-derangement numbers $d_n(q)$ satisfy the recurrence*

- $d_0(q) = 1$;
- $d_n(q) = [n]_q d_{n-1}(q) + (-1)^n q^{\binom{n}{2}}$, for $n \geq 1$.

For a permutation $\sigma \in \mathfrak{S}_n$, its *reduced major index*, introduced by Désarménien [Dés94], is the unique integer $\text{maj}^*(\sigma) \in \{0, 1, \dots, n-1\}$ satisfying

$$\text{maj}^*(\sigma) \equiv \text{maj}(\sigma) \pmod{n}. \quad (1.6)$$

Using the recurrence for the q -derangement numbers, Désarménien obtained an almost uniform distribution of the reduced major index over derangements, see (1.12). For $0 \leq r \leq n-1$, set

$$d_n^r = |\{\sigma \in \mathcal{D}_n : \text{maj}^*(\sigma) = r\}|. \quad (1.7)$$

Let $d_n^*(q)$ denote the remainder of $d_n(q)$ upon division by $1 - q^n$. It follows that

$$d_n^*(q) = \sum_{r=0}^{n-1} d_n^r q^r.$$

Lemma 1.3. *If $P(q)$ is a polynomial in q , then we have*

$$P(q)[n]_q \equiv P(1)[n]_q \pmod{1 - q^n}. \quad (1.8)$$

Proof. Since $P(q) - P(1)$ vanishes at $q = 1$, there exists a polynomial $Q(q)$ such that

$$P(q) - P(1) = (1 - q)Q(q).$$

Therefore, we have

$$P(q)[n]_q - P(1)[n]_q = (1 - q)Q(q) \frac{1 - q^n}{1 - q} = (1 - q^n)Q(q). \quad (1.9)$$

Eq. (1.8) immediately follows from the above identity. $\square$

Applying Lemma 1.3 to $P(q) = d_{n-1}(q)$, we obtain

$$d_{n-1}(q)[n]_q \equiv d_{n-1}(1)[n]_q \pmod{1 - q^n}. \quad (1.10a)$$

Moreover, it is easy to check that

$$(-1)^n q^{\binom{n}{2}} \equiv \begin{cases} q^{n/2}, & \text{if } n \text{ is even;} \\ -1, & \text{if } n \text{ is odd;} \end{cases} \pmod{1 - q^n}. \quad (1.10b)$$

Indeed, if n is even, then $\binom{n}{2} \equiv n/2 \pmod{n}$ and $(-1)^n = 1$; whereas if n is odd, then $\binom{n}{2} \equiv 0 \pmod{n}$ and $(-1)^n = -1$. Combining these congruences (1.10) with the recurrence in Proposition 1.2 yields Désarménien's formula.

Theorem 1.4 ([Dés94, Proposition 2.2]). *For $n \geq 1$, we have*

$$d_n^*(q) = d_{n-1}[n]_q + \begin{cases} q^{n/2}, & n \text{ is even;} \\ -1, & n \text{ is odd.} \end{cases} \quad (1.11)$$

Equivalently, for $0 \leq r \leq n - 1$, we have

$$d_n^r = \begin{cases} d_{n-1} + 1, & n \text{ is even and } r = n/2; \\ d_{n-1} - 1, & n \text{ is odd and } r = 0; \\ d_{n-1}, & \text{otherwise.} \end{cases} \quad (1.12)$$

Summing (1.12) over $0 \leq r \leq n - 1$, we recover the recurrence (1.2). At the end of his paper, Désarménien [Dés94] asked for a combinatorial proof of his formula. The purpose of this paper is to provide a combinatorial interpretation of Désarménien's formula via decorated permutations.

2. A COMBINATORIAL INTERPRETATION OF THEOREM 1.4

We can extend the descent set and the major index from ordinary permutations to decorated permutations in E_n using the following total order

$$\bar{n} < \dots < \bar{2} < \bar{1} < 1 < 2 < \dots < n. \quad (2.1)$$

Example 1. *If $\sigma = \bar{3}2\bar{6}514$, then $\text{Des}(\sigma) = \{2, 4\}$ and $\text{maj}(\sigma) = 6$.*

Let $G_n := E_n \setminus \mathcal{D}_n$ be the set of decorated permutations on $[n]$ with at least one fixed point, either signed or unsigned. For $\sigma \in E_n$, denote by $\text{nfix}(\sigma)$ the number of signed fixed points of σ . The following cancelation identity is an immediate consequence of Ji's sign-reversing and descent-set-preserving involution on G_n .

Theorem 2.1 ([Ji26]). *We have*

$$\sum_{\sigma \in E_n} (-1)^{\text{nfix}(\sigma)} q^{\text{maj}(\sigma)} = d_n(q). \quad (2.2)$$

Let π and σ be two words with distinct letters. A *shuffle* of σ and π is a word obtained by interleaving their letters while preserving the relative order of the letters within each word. We denote the set of all such shuffles by $\pi \sqcup \sigma$.

Example 2. *If $\pi = 13$ and $\sigma = 24$, then we have*

$$13 \sqcup 24 = \{1324, 1234, 1243, 2134, 2143, 2413\}.$$

For $0 \leq k \leq n$, define the subset of E_n

$$E_{n,k} = \{\sigma \in E_n : \text{nfix}(\sigma) = k\}. \quad (2.3)$$

Lemma 2.2. *Let $\delta_k = k(k-1) \cdots 1$. There is a descent-set-preserving bijection¹*

$$\phi_{n,k} : E_{n,k} \longrightarrow \bigsqcup_{\beta \in \mathfrak{S}_{\{k+1, \dots, n\}}} \delta_k \sqcup \beta, \quad (2.4)$$

where $\mathfrak{S}_{\{k+1, \dots, n\}}$ denotes the set of permutations of $\{k+1, \dots, n\}$.

Proof. For $\pi \in E_{n,k}$, let $i_1 < i_2 < \cdots < i_k$ be the positions of its signed fixed points. Thus, we have $\pi_{i_j} = \overline{i_j}$ for $(1 \leq j \leq k)$. Define $\phi_{n,k}(\pi)$ as follows. Replace the signed fixed points $\overline{i_1}, \overline{i_2}, \dots, \overline{i_k}$ from left to right by $k, k-1, \dots, 1$, respectively. Then standardize the remaining positive letters by replacing the j -th smallest one by $k+j$. The resulting word is a permutation of $[n]$. Moreover, the subword formed by the letters $1, 2, \dots, k$ is $\delta_k := k(k-1) \cdots 1$, while the remaining letters form a permutation $\beta \in \mathfrak{S}_{\{k+1, \dots, n\}}$. Hence $\phi_{n,k}(\pi) \in \delta_k \sqcup \beta$.

Conversely, let $w \in \delta_k \sqcup \beta$ for some $\beta \in \mathfrak{S}_{\{k+1, \dots, n\}}$. Let $i_1 < i_2 < \cdots < i_k$ be the positions occupied in w by the letters of δ_k . Replace the letter $k-j+1$ at position i_j by the signed fixed point $\overline{i_j}$. Let $a_1 < a_2 < \cdots < a_{n-k}$ be the elements of $[n] \setminus \{i_1, \dots, i_k\}$. For each remaining letter $k+j$ of w , replace it by a_j . This uniquely reconstructs a decorated permutation $\pi \in E_{n,k}$. Thus $\phi_{n,k}$ is a bijection.

It remains to verify that $\phi_{n,k}$ preserves descent sets. If two adjacent letters of π are both positive, their relative order is preserved by standardization. If they are both signed fixed points, then their absolute values increase from left to right, whereas the order $\overline{n} < \cdots < \overline{2} < \overline{1}$ makes them decreasing; their images also form a decreasing subword of δ_k . Finally, every signed fixed point is smaller than every positive letter, and its image belongs to $\{1, \dots, k\}$, while the image of every positive letter belongs to $\{k+1, \dots, n\}$. Thus, all adjacent comparisons are preserved, and hence $\text{Des}(\phi_{n,k}(\pi)) = \text{Des}(\pi)$. $\square$

Insertion families. Fix $0 \leq k < n$, and let $\pi \in E_{n,k}$ and the corresponding permutation $w = \phi_{n,k}(\pi)$. Since $k < n$, the largest letter n does not belong to the subword δ_k . Delete n from w , and denote the resulting permutation by w' . Then $w' \in \mathfrak{S}_{n-1}$ and $w'|_{\{1, \dots, k\}} = \delta_k$. (Here $w'|_{\{1, \dots, k\}}$ denotes the subword of w' obtained by retaining only the letters $1, \dots, k$.) Hence, there is a unique permutation $\pi' \in E_{n-1,k}$ such that $w' = \phi_{n-1,k}(\pi')$.

¹This canonical bijection appears in Ji's proof. Since its explicit construction is not given there, we provide the details for completeness.

Conversely, starting with $\pi' \in E_{n-1,k}$, one may insert the largest letter n into any of the n gaps of $w' = \phi_{n-1,k}(\pi')$. The resulting permutation still contains δ_k as the prescribed subword. Thus, at the level of sets, we obtain a bijection

$$\psi : E_{n-1,k} \times \{0, 1, \dots, n-1\} \longrightarrow E_{n,k}. \quad (2.5)$$

To control the major index, we use the standard major index insertion labeling, which is the so-called Carlitz's insertion method; see [Car75; Wil16].

Lemma 2.3. *Let $w' = w'_1 w'_2 \cdots w'_{n-1} \in \mathfrak{S}_{n-1}$ with $\text{des}(w') = d$. Label the n gaps of w' according to the following rule:*

- label the gap after w'_{n-1} by 0;
- label the gaps immediately following the descents of w' from right to left by $1, 2, \dots, d$;
- label the gap before w'_1 by $d+1$;
- label the gaps immediately following the ascents of w' from left to right by $d+2, d+3, \dots, n-1$.

If letter n is inserted into the gap labeled r , then the resulting permutation $w \in \mathfrak{S}_n$ satisfies

$$\text{maj}(w) = \text{maj}(w') + r.$$

Example 3. If $\sigma = 5167324$, then it receives the following labels,

$$\begin{array}{cccccccc} 5 & 1 & 6 & 7 & 3 & 2 & 4 & \\ 4 & 3 & 5 & 6 & 2 & 1 & 7 & 0 \end{array}.$$

For each $\pi' \in E_{n-1,k}$, the bijection ψ associates an insertion family consisting of n elements of $E_{n,k}$. By Lemma 2.3, their major indices are

$$\text{maj}(\pi'), \text{maj}(\pi') + 1, \dots, \text{maj}(\pi') + n - 1.$$

Since the number of signed fixed points remains equal to k , we obtain

$$\sum_{\pi \in E_{n,k}} (-1)^k q^{\text{maj}(\pi)} = \sum_{\pi' \in E_{n-1,k}} \sum_{r=0}^{n-1} (-1)^k q^{\text{maj}(\pi') + r} = [n]_q \sum_{\pi' \in E_{n-1,k}} (-1)^k q^{\text{maj}(\pi')}. \quad (2.6)$$

The exceptional object. When $k = n$, the set $E_{n,n}$ consists of a single decorated permutation, $\omega_n = \bar{1} \bar{2} \cdots \bar{n}$. Since $\bar{1} > \bar{2} > \cdots > \bar{n}$, we have $\text{Des}(\omega_n) = \{1, 2, \dots, n-1\}$ and hence

$$\text{maj}(\omega_n) = 1 + 2 + \cdots + (n-1) = \binom{n}{2}.$$

Moreover, $(-1)^{\text{nfix}(\omega_n)} = (-1)^n$. Thus, the signed weight of ω_n is $(-1)^n q^{\binom{n}{2}}$.

Summing (2.6) over $0 \leq k < n$ and then adding the contribution of ω_n , we obtain

$$\sum_{\pi \in E_n} (-1)^{\text{nfix}(\pi)} q^{\text{maj}(\pi)} = [n]_q \sum_{\pi' \in E_{n-1}} (-1)^{\text{nfix}(\pi')} q^{\text{maj}(\pi')} + (-1)^n q^{\binom{n}{2}}. \quad (2.7)$$

Together with Theorem 2.1, this identity recovers the recurrence

$$d_n(q) = [n]_q d_{n-1}(q) + (-1)^n q^{\binom{n}{2}}.$$

We next refine the same insertion decomposition by recording the reduced major index.

Proof of Désarménien's formula. Consider the signed reduced-major-index generating polynomial

$$F_n(q) := \sum_{\pi \in E_n} (-1)^{\text{fix}(\pi)} q^{\text{maj}^*(\pi)}.$$

Since Ji's involution is sign-reversing and preserves the major index, it also preserves the reduced major index. Thus all decorated permutations in $G_n = E_n \setminus \mathcal{D}_n$ cancel in pairs. The uncanceled objects are precisely the ordinary derangements in $\mathcal{D}_n$ and hence

$$F_n(q) = \sum_{\sigma \in \mathcal{D}_n} q^{\text{maj}^*(\sigma)} = d_n^*(q). \quad (2.8)$$

For each $\pi' \in E_{n-1,k}$, where $0 \leq k < n$, let

$$\mathcal{I}(\pi') := \{\psi(r, \pi') : 0 \leq r \leq n-1\}$$

be its insertion family. Combining Lemma 2.3 with the fact that $\phi_{n,k}$ and $\phi_{n-1,k}$ preserve the major index, the elements of $\mathcal{I}(\pi')$ have major indices

$$\text{maj}(\pi') + r, \quad 0 \leq r \leq n-1.$$

Their residues modulo n run through $0, 1, \dots, n-1$, each exactly once. Since all elements of $\mathcal{I}(\pi')$ have sign $(-1)^{\text{fix}(\pi')}$, this family contributes $(-1)^{\text{fix}(\pi')} [n]_q$ to $F_n(q)$.

Let ι denote Ji's involution on G_{n-1} . If $\pi' \in G_{n-1}$, then $(-1)^{\text{fix}(\iota(\pi'))} = -(-1)^{\text{fix}(\pi')}$. Consequently, the two insertion families $\mathcal{I}(\pi')$ and $\mathcal{I}(\iota(\pi'))$ have opposite contributions and cancel. Thus, only the families indexed by $\pi' \in \mathcal{D}_{n-1}$ remain. Since $|\mathcal{D}_{n-1}| = d_{n-1}$, the total contribution of the insertion families is $d_{n-1}[n]_q$.

It remains to account for the unique element of $E_{n,n}$, namely $\omega_n = \bar{1}\bar{2}\cdots\bar{n}$. Its signed weight is $(-1)^n q^{\binom{n}{2} \bmod n}$. Since

$$\binom{n}{2} \equiv \begin{cases} n/2 \pmod{n}, & \text{if } n \text{ is even;} \\ 0 \pmod{n}, & \text{if } n \text{ is odd,} \end{cases}$$

its contribution to $F_n(q)$ is

$$\begin{cases} q^{n/2}, & \text{if } n \text{ is even;} \\ -1, & \text{if } n \text{ is odd.} \end{cases}$$

Combining the two contributions, we obtain

$$d_n^*(q) = F_n(q) = d_{n-1}[n]_q + \begin{cases} q^{n/2}, & \text{if } n \text{ is even;} \\ -1, & \text{if } n \text{ is odd.} \end{cases}$$

This completes the proof. □

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REFERENCES

- [BO17] A. T. Benjamin and J. Ornstein. “A bijective proof of a derangement recurrence”. *Fibonacci Quart.* 55.5 (2017), pp. 28–29 (cit. on p. 1).
- [Car75] L. Carlitz. “A combinatorial property of q -Eulerian numbers”. *Amer. Math. Monthly* 82 (1975), pp. 51–54 (cit. on p. 5).
- [Dés84] J. Désarménien. “Une autre interprétation du nombre de dérangements”. *Sém. Lotharing. Combin.* 9 (1984), pp. 11–16 (cit. on p. 1).
- [Dés94] J. Désarménien. “Distribution de l’indice majeur réduit sur les dérangements”. *Sém. Lothar. Combin.* 32 (1994), Art. B32a, approx. 3 (cit. on pp. 2, 3).
- [Eli21] S. Elizalde. “A simple bijective proof of a familiar derangement recurrence”. *Fibonacci Quart.* 59.2 (2021), pp. 150–151 (cit. on p. 1).
- [GR93] I. M. Gessel and C. Reutenauer. “Counting permutations with given cycle structure and descent set”. *J. Combin. Theory Ser. A* 64.2 (1993), pp. 189–215 (cit. on p. 2).
- [Ji26] K. Q. Ji. *Revisiting q -Derangement Numbers via Decorated Permutations*. 2026. arXiv: 2607.04798 [math.CO] (cit. on pp. 2, 4).
- [Mac60] P. A. MacMahon. *Combinatory analysis*. Two volumes (bound as one). Chelsea Publishing Co., New York, 1960, pp. xix+302+xix+340 (cit. on p. 1).
- [Pos06] A. Postnikov. “Total positivity, Grassmannians, and networks”. *arXiv preprint math/0609764* (2006) (cit. on p. 2).
- [Rem83] J. B. Remmel. “A note on a recursion for the number of derangements”. *European J. Combin.* 4.4 (1983), pp. 371–374 (cit. on p. 1).
- [Wac89] M. L. Wachs. “On q -derangement numbers”. *Proc. Amer. Math. Soc.* 106.1 (1989), pp. 273–278 (cit. on p. 2).
- [Wil84] H. S. Wilf. “A Bijection in the Theory of Derangements”. *Mathematics Magazine* 57.1 (1984), pp. 37–40 (cit. on p. 1).
- [Wil16] A. T. Wilson. “An extension of MacMahon’s equidistribution theorem to ordered multiset partitions”. *Electron. J. Combin.* 23.1 (2016), Paper 1.5, 21 (cit. on p. 5).

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